Evaluating the Robustness of Large Language Models for Image Captioning on Adversarially Attacked Images

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April 2025

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES

(A.P. Government Act 18 of 2008) RGUKT-NUZVID,
Krishna Dist - 521202

CERTIFICATE OF COMPLETION

This is to certify that the work entitled **"Evaluating the Robustness of Large Language Models for Image Captioning on Adversarially Attacked Images"** is the bonafide work of:

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carried out under my guidance for the Minor Project of the Bachelor of Technology during the academic session December 2024 to April 2025.

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CERTIFICATE OF EXAMINATION

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and here by accord our approval of it as a study carried out and presented in a manner required for its acceptance in the 3rd year of Bachelor Of Technology for which it has submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn, as recorder in this thesis. It only signifies the acceptance of this thesis for the purpose for which it has been submitted.

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DECLARATION

We, **V. Madhu Sree (N200806), K. Swathi (N200774), P. N.P. Asritha (N200621), G. Sathish Roy (N200298), and N. Abhiram Naidu (N200211)**, hereby declare that the project report entitled "Evaluating the Robustness of Large Language Models for Image Captioning on Adversarially Attacked Images" done by us under the guidance of Dr. S. Chiranjeevi is submitted in complete fulfillment for the Minor Project of Bachelor of Technology in Computer Science and Engineering during the academic session December 2024 to April 2025 at RGUKT-Nuzvid.

We also declare that this project is a result of our own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

Place: Nuzvid
Date:

ACKNOWLEDGEMENT

I would like to express my gratitude towards my advisor Dr.Sadu Chiranjeevi sir for guiding me through the extensive research process. I would also like to thank my team members for their support. Lastly, I would like to thank my family and friends for their encouragement, suggestions and belief in me.

We are extremely grateful for the confidence bestowed in us and entrusting our project entitled "Evaluating the Robustness of Large Language Models for Image Captioning on Adversarially Attacked Images".

We express our gratitude to S.Bhavani mam (HOD of CSE) and other faculty members for being a source of inspiration and constant encouragement, which helped us in completing the project successfully.

Finally, yet importantly, we would like to express our heartfelt thanks to our beloved God and parents for their blessings, and our friends for their help and wishes for the successful completion of this project

ABSTRACT

Image captioning models, which generate text descriptions for images, have improved a lot with the use of large language models (LLMs). However, these models can still be easily fooled by small changes in the images that are not visible to the human eye. In this project, we study how well popular image captioning LLMs handle images that have been attacked with such hidden changes, called adversarial attacks. We test the models on both normal and attacked images and compare the captions they produce. Our results show how these small changes can confuse the models and affect the quality of their captions. This project highlights the Robustness of the LLM models.

Contents

Chapter 1

Introduction

1.1 Introduction to project

Image captioning is the process of automatically generating text descriptions for images using AI models. Large Language Models (LLMs) have made this task more accurate and natural. However, these models can be fooled by small changes in the images—called adversarial attacks—that are not noticeable to humans but can confuse the models.
This project focuses on testing how well popular image captioning models like BLIP-2, GIT, and ViT-GPT-2 perform when such attacks are applied. By comparing the captions from original and attacked images, we aim to understand how robust these models are and how much their performance drops under different types of adversarial conditions.
This project explores how robust these image captioning LLMs are when subjected to such adversarial perturbations. By applying attacks like FGSM, PGD, DeepFool, AutoAttack, and others, and evaluating performance using metrics such as BLEU, METEOR, and BERTScore, we aim to highlight the vulnerabilities in existing models and the importance of building more resilient AI systems for deployment in sensitive and critical environments.

1.2 Objectives

- To test how well image captioning models work when images are slightly changed using tricks called adversarial attacks.
- To compare the captions from normal images and attacked images
- To show why it's important to make these models more secure for real-world use.

1.3 Motivation

Image captioning models have become an important part of many AI applications by helping machines understand and describe images in natural language. While large language models (LLMs) have greatly improved caption quality, they are still vulnerable to adversarial attacks — small, hidden changes to images that can trick models into making mistakes

1.4 Adversarial attacks

Adversarial attacks refer to techniques used to trick machine learning models into making incorrect predictions or causing unintended behavior.

1.5 Consequences

- Wrong Descriptions - The models may generate incorrect captions that do not match the image, which can mislead users.
- Safety Risks - In real-world applications like self-driving cars or medical tools, wrong captions can cause dangerous decisions.
- Security Issues - Attackers can use these tricks to fool systems on purpose, leading to misuse or harm.
- Reduced Accuracy - The overall performance of the model drops, especially in important tasks like content moderation or assistive tools for the visually impaired.

1.6 Real-World Applications

- Autonomous Vehicles: Tiny changes to road signs (like adding stickers or graffiti) can trick the car's vision system into misclassifying stop signs as speed limits.
- Medical Diagnosis Systems: Slight alterations to medical images (X-rays, MRIs) can cause AI systems to misdiagnose diseases.
- Assistive Technology: Image captioning helps visually impaired users by describing scenes to them. Incorrect captions caused by attacks could mislead users and cause safety risks.
- Content Moderation: Platforms like social media use captioning models to automatically detect and tag inappropriate or sensitive content. Adversarial attacks can cause wrong tagging or missed detections.

Chapter 2

System Requirements

2.1 Hardware

- **Processor**:Cloud-based (Google Colab environment, backend CPU: Intel Xeon or equivalent)
- **RAM** : 12 GB or more (Colab GPU instance)
- **GPU**: NVIDIA Tesla T4/P100/V100 (provided by Colab)
- **Storage**: 50 GB (Cloud Storage and temporary runtime disk space)

2.2 Software

- **OS**: Ubuntu/Linux (Google Colab runtime environment)
- **Python** 3.11.4
- **Libraries:**
  - Flask (for minimal API if needed)
  - torch (PyTorch for deep learning models)
  - torchvision (for image transformations and pre-trained models)
  - NumPy (for numerical computations)
  - Pandas (for dataset handling)
  - scikit-learn (for evaluation metrics)
  - transformers (for using image captioning LLMs)
  - Pillow (for image processing)
  - Matplotlib (for visualization)
  - Adversarial Robustness Toolbox (ART) (for generating adversarial attacks)

2.3 Functional Requirements

- Caption Generation
- Generate captions for clean and adversarially attacked images using various large language models.
- Adversarial Attack Generation.
- Apply different adversarial attack techniques (like FGSM, PGD) to test the models robustness.
- Robustness Evaluation
- Compare arid analyze the change in caption quality before and after attacks using metrics like BLEU, ROUGE, METEOR
- Visualization and Analysis
- Visualize attacked images and their captions to qualitatively assess model performance.

2.4 Non-Functional Requirements

- Jupyter Notebook interface on Google Colab for easy experimentation and visualization.
- Compatibility
- Security
- Performance
- Private datasets and model access, no exposure of API keys or confidential resources.
- Efficient use of GPU resources, optimized data loading and model evaluation for faster experimentation.

Chapter 3

System Design

3.1 Flowchart

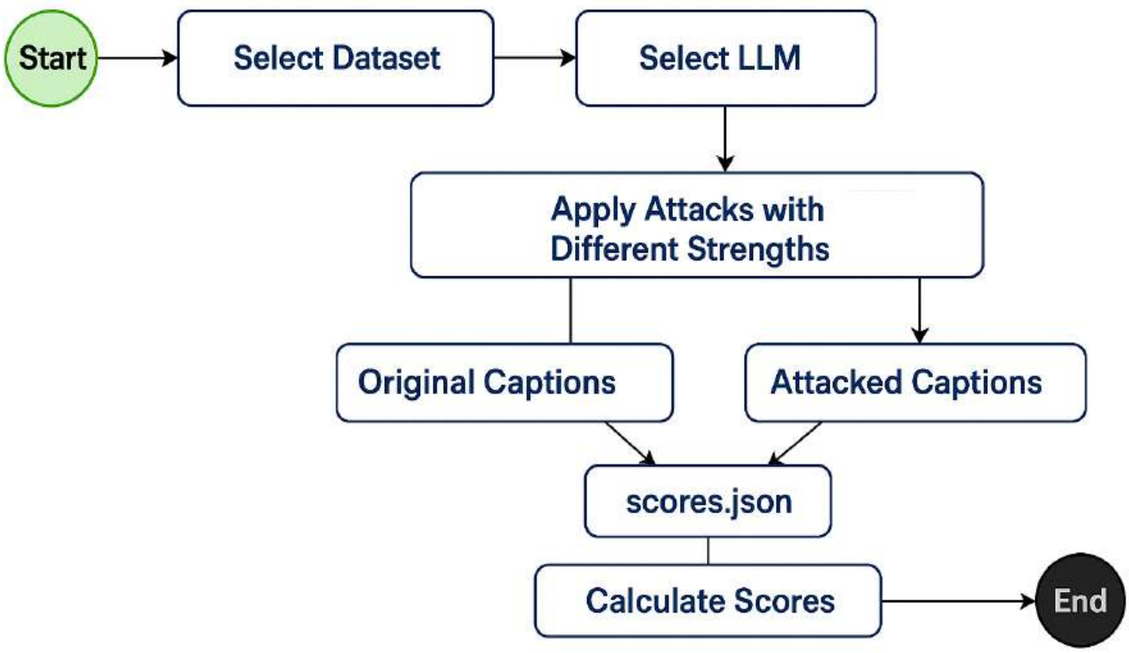


Figure 3.1: Overall Workflow of the Proposed System

3.2 Component Explanation

- **Select Dataset**
  - **Meaning**: Choose the dataset with images and their captions that we want to use.

- **Purpose**: The selected dataset provides the base images on which captioning and attacks will happen.

- **SelectLLM**
  - **Meaning:** Select the Large Language Model (LLM) that will generate captions for the images.
  - **Purpose**: The quality and type of captions depend on the LLM you choose (e.g., BLIP-2, GIT, Pali, VIT Gpt-2 etc.).

- **Apply Attacks with Different Strengths**
  - **Meaning**: Adversarial attacks (small perturbations to images) are applied at various levels of intensity (different strengths).
  - **Purpose:** To test how robust the LLM is when images are slightly altered (e.g.,FGSM, PGD, Autoattack, Deepfool, Bim etc.)

- **Original Captions**
  - **Meaning:** Captions generated by the LLM on the clean/original images without any attacks.
  - **Purpose:** They act as a reference or baseline to compare with captions from attacked images.

- **Attacked Captions**
  - **Meaning:** Captions generated by the LLM on the attacked/perturbed images.
  - **Purpose:** To see how the LLM's captioning changes (or fails) when faced with adversarial inputs.

- **Scores.json**
  - **Meaning**: A JSON file storing both original and attacked captions, ground truth,and possibly the image IDs.
  - **Purpose**: To organize and save the data needed for calculating evaluation metrics.

- **Calculate Scores**
  - **Meaning:** Scores are calculated to measure how much captions changed after attacks like BLEU, ROUGE, METEOR, CIDEr.
  - **Purpose:** To quantify the impact of attacks and evaluate the robustness of the LLM.

3.3 Algorithm

- **Step 1**: Start
  Begin the robustness evaluation workflow.

- **Step 2:** Select Dataset
Choose a dataset containing text samples (e.g., image captions, prompts, etc.) to be used for evaluation.

- **Step 3:** Select LLM
Choose a large language model (LLM) to test (e.g., GPT, LLaMA, etc.).

- **Step 4**: Apply Adversarial Attacks
Apply adversarial attacks on the selected dataset using the chosen LLM. Vary the strength/intensity of the attacks. Generate perturbed (attacked) versions of the original samples.

- **Step 5**: Generate Captions
Use the LLM to generate:
Original Captions: Outputs from clean (unaltered) inputs.
Attacked Captions: Outputs from adversarially perturbed inputs.

- **Step 6**: Save Captions
Save both sets of captions (original and attacked) along with metadata in a file (e.g., scores.json).

- **Step 7**: Calculate Evaluation Scores
Evaluate the impact of the attacks by comparing original and attacked captions using selected metrics (e.g., BLEU, ROUGE, semantic similarity). Store results for analysis.

- **Step 8** End
Complete the evaluation process. Results can now be used to assess LLM robustness.

3.4 Experimental Setup

3.4.1 BLIP-2

- Technologies/Libraries Used
  - Deep Learning Frameworks: PyTorch, Transformers, Hugging Face Hub
  - Computer Vision and Image Processing: Torchvision, Pillow (PIL), Matplotlib
  - Adversarial Attacks: FGSM, PGD, Deepfool, AutoAttack, c&w
  - Evaluation Metrics: NLTK, pycocoevalcap (METEOR, CIDEr), BERTScore
  - Utility Libraries: TQDM, NumPy, gc, datetime
- Datasets
  - Flicker8k
- System Hardware
  - Google Colab (GPU Environment)

- Results
  - The evaluation of various attack methods on the performance of image captioning models is presented below.

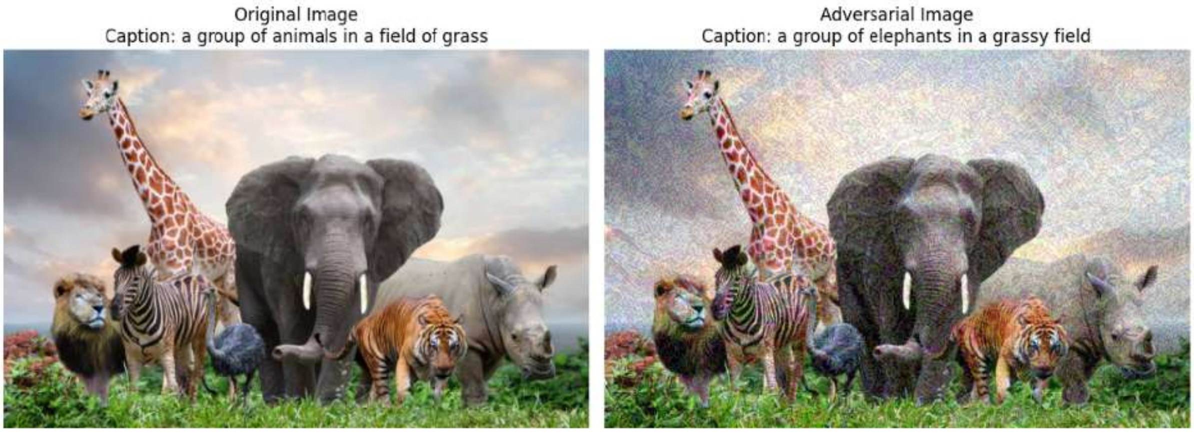


Figure 3.2: Impact of FGSM

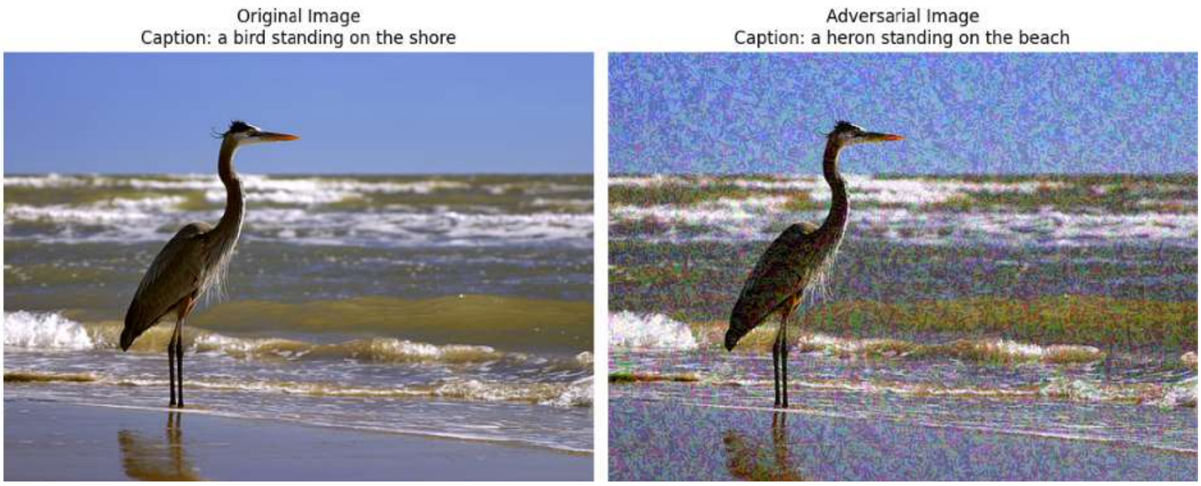


Figure 3.3: Impact of PGD

Figure 3.4: Impact of C&W

Figure 3.5: Impact of Deepfool

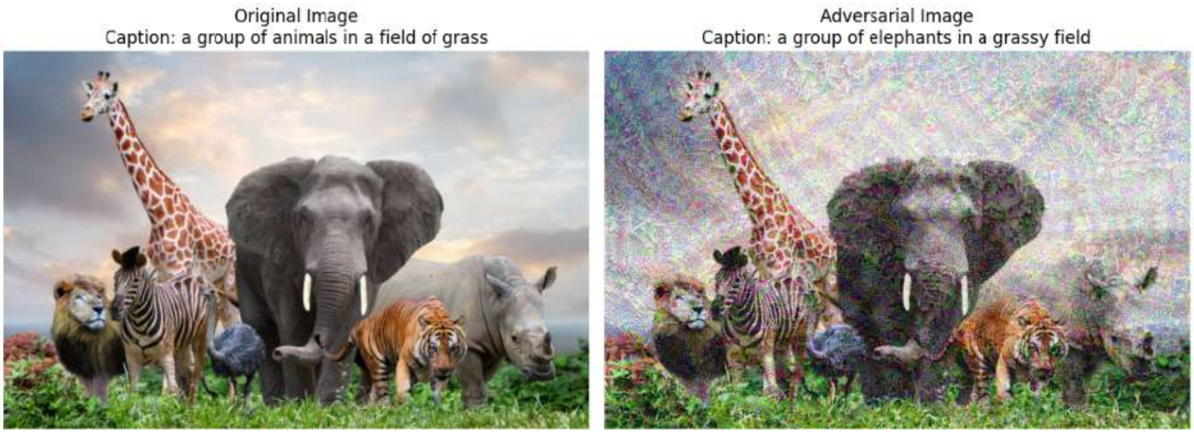


Figure 3.6: Impact of Autoattack

- The following tables display the impact of different adversarial attacks on key metrics such as BLEU, METEOR, and BERTScore F1. Each attack is accompanied by its strength parameters, and the corresponding drop in performance for each metric is shown to demonstrate the severity of each attack.

Table 3.1: Evaluation of Attack Methods on BLIP-2

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.2264	-	0.2391	-	-
FGSM	$\epsilon = 0.1$	0.0891	0.1373	0.1132	0.1259	0.9059
PGD	$\epsilon = 0.2$, $\alpha = 0.2$, iters=5	0.2066	0.0198	0.2245	0.0146	0.9570
C&W	$c = 1e{-3}$, lr=0.05, iters=200	0.1986	0.0278	0.2186	0.0205	0.9562
DeepFool	iters=150, overshoot=0.08	0.2012	0.0252	0.2166	0.0225	0.9512
AutoAttack	$\epsilon = 0.5$	0.1078	0.1186	0.1268	0.1123	0.9101

Table 3.2: Evaluation of Attack Methods on BLIP-2

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.2264	-	0.2391	-	-
FGSM	$\epsilon = 0.2$	0.0800	0.1464	0.1100	0.1291	0.8900
PGD	$\epsilon = 0.4$, $\alpha = 0.4$, num_iters=10	0.1700	0.0564	0.1900	0.0491	0.9200
C&W	$c = 1e{-}2$, lr=0.1, num_iters=300	0.1600	0.0664	0.1800	0.0591	0.9300
DeepFool	num_iters=200, overshoot=0.1	0.1700	0.0564	0.1900	0.0491	0.9200
AutoAttack	$\epsilon = 0.7$	0.0900	0.1364	0.1200	0.1191	0.8800

- Results Comparison with Other Approaches
  - As attack strength increases, BLIP-2's performance worsens significantly, especially under FGSM and AutoAttack, showing the model's vulnerability and the need for better defense mechanisms

3.4.2 GIT

- Technologies/Libraries Used
  - Deep Learning Frameworks: PyTorch, Transformers
  - Computer Vision and Image Processing: Torchvision, Pillow (PIL)
  - Adversarial Attack Libraries: FGSM, PGD, CW, JSMA, One Pixel Attack
  - Evaluation Metrics: NLTK, pycocoevalcap (METEOR, CIDEr), BERTScore
  - Utility Libraries: TQDM, NumPy
- Datasets
  - Flicker8k
- System Hardware
  - Google colab(GPU Environment)
- Results
  - The evaluation of various attack methods on the performance of image captioning models is presented below.

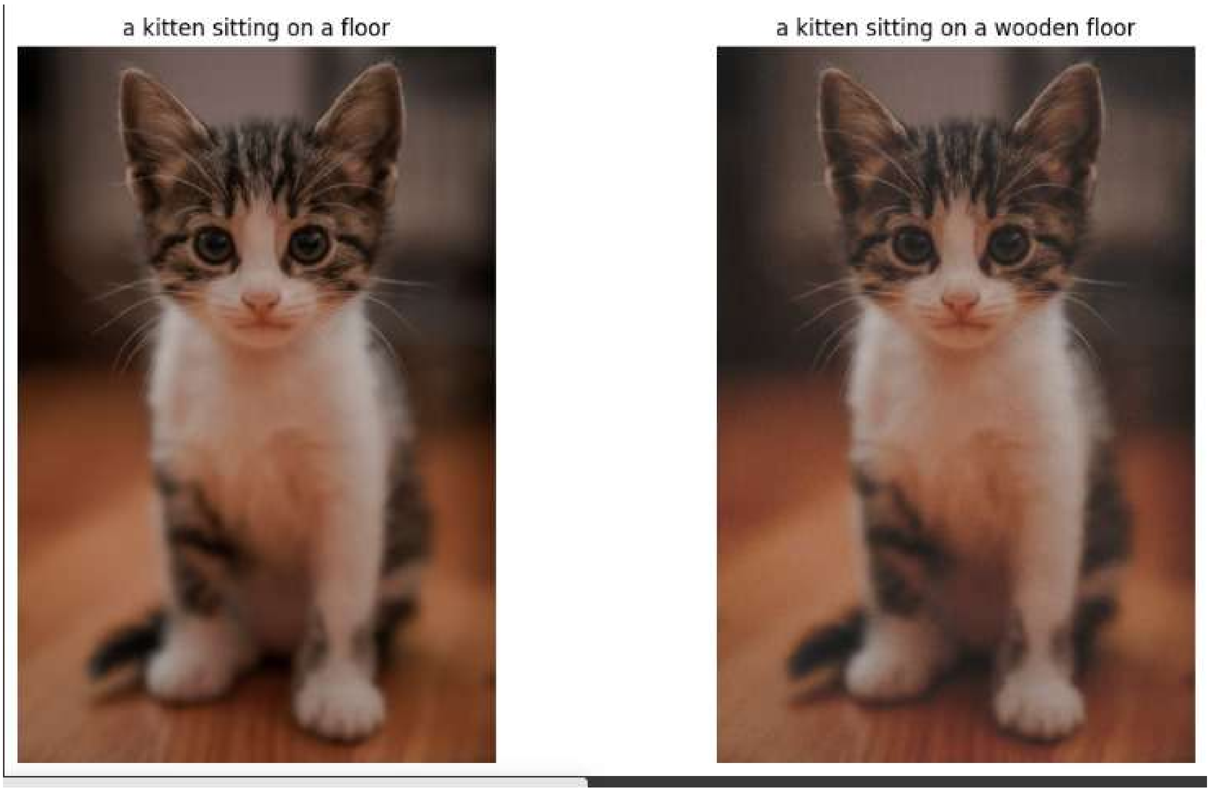


Figure 3.7: Impact of PGD

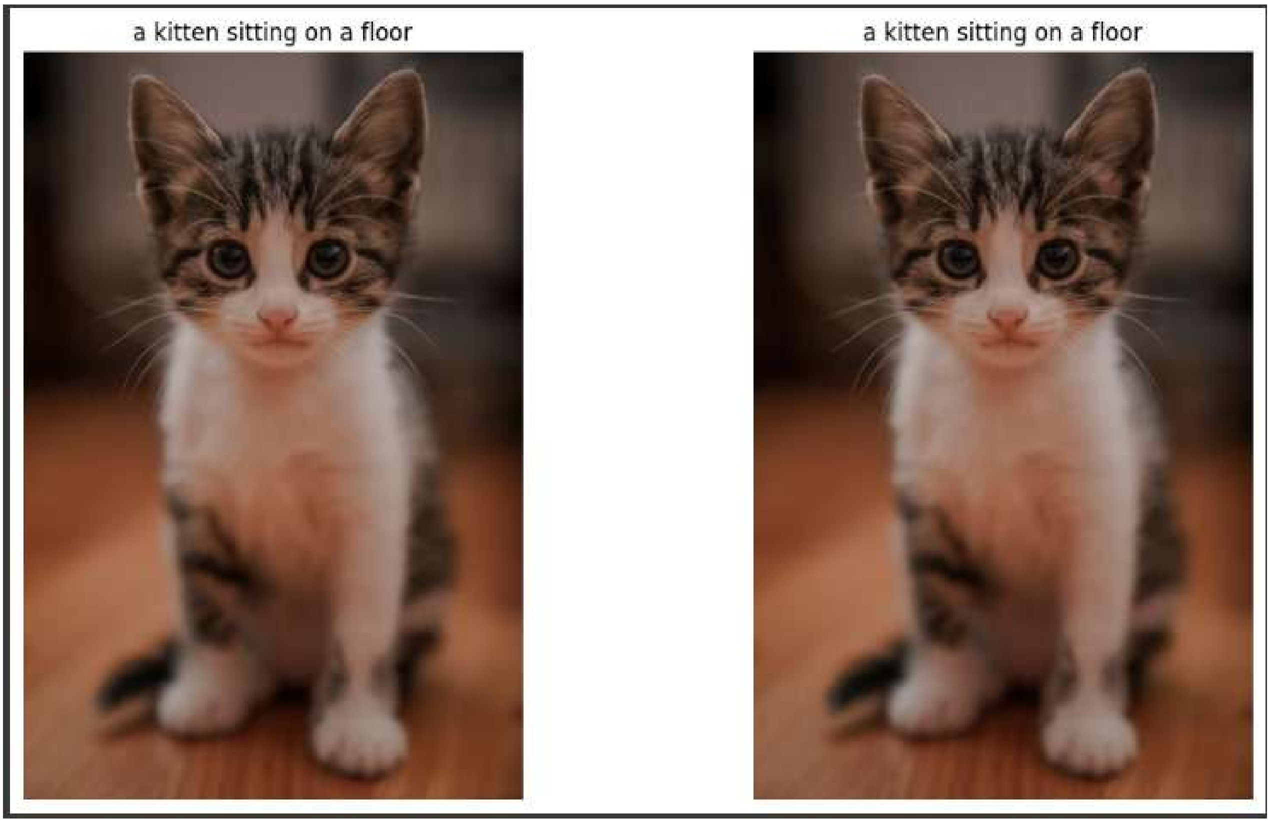


Figure 3.8: Impact of FGSM

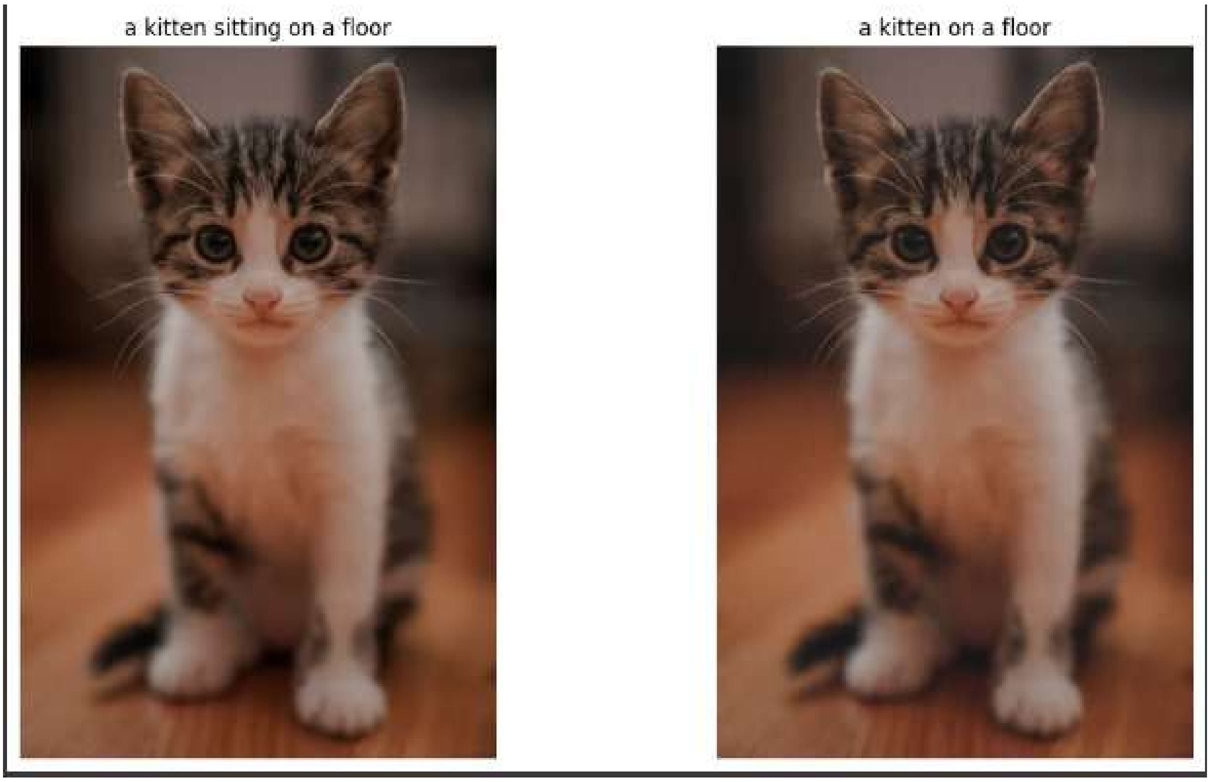


Figure 3.9: Impact of C&W

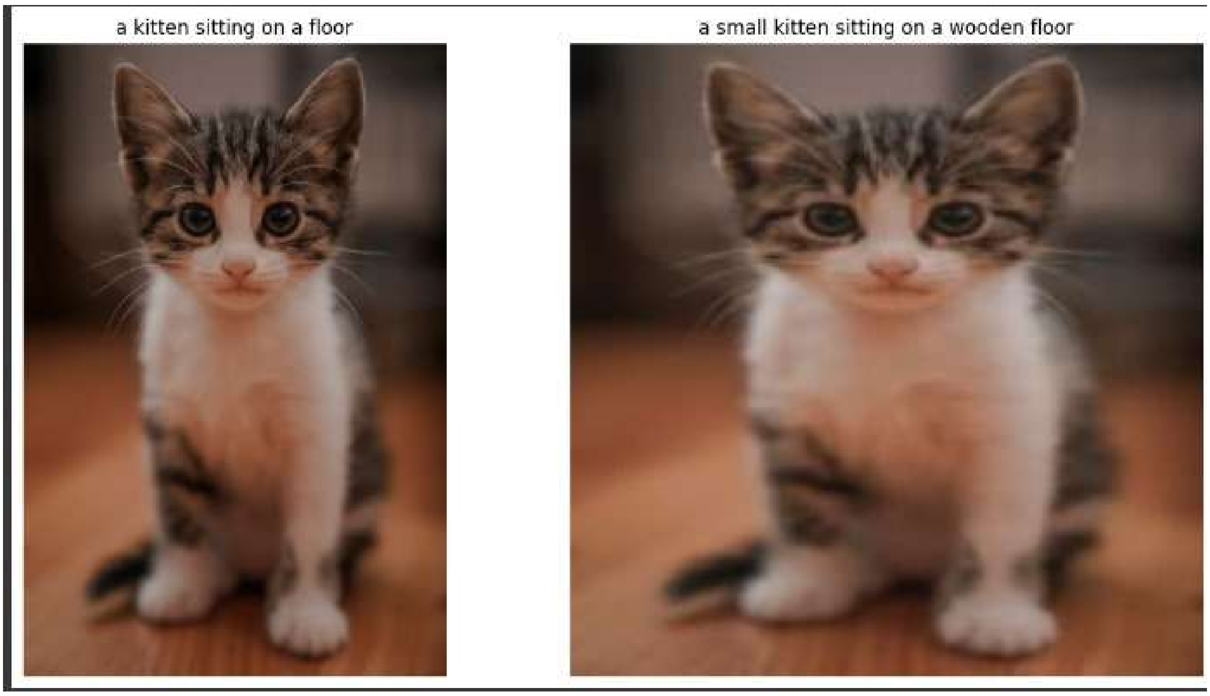


Figure 3.10: Impact of JSMA

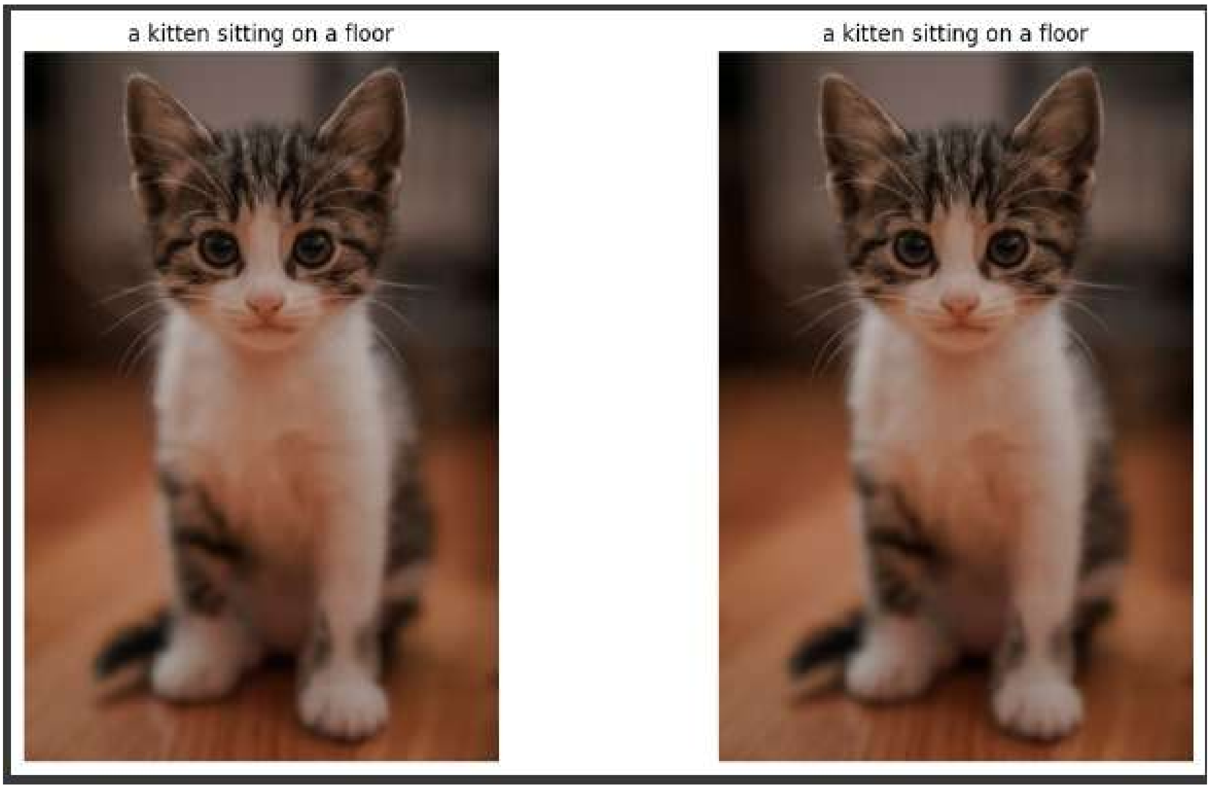


Figure 3.11: Impact of One Pixel

- The following tables display the impact of different adversarial attacks on key metrics such as BLEU, METEOR, and BERTScore F1. Each attack is accompanied by its strength parameters, and the corresponding drop in performance for each metric is shown to demonstrate the severity of each attack.

Table 3.3: Evaluation of Attack Methods on GIT

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1528	-	0.1815	-	-
FGSM	eps=0.02	0.1535	-0.0007	0.1824	-0.0009	0.9926
PGD	eps=0.2, α=0.01, num_iters=1	0.1487	0.0041	0.1795	0.002	0.9749
C&W	eps=0.03	0.1266	0.0262	0.1702	0.0113	0.9184
JSMA	eps=0.05	0.1529	-0.0001	0.1819	-0.0004	0.9862
One Pixel Attack	num_pixels=1	0.1537	-0.0009	0.1815	0.0	0.996

Table 3.4: Evaluation of Attack Methods on GIT

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1528	-	0.1815	-	-
FGSM	$\epsilon = 0.1$	0.1200	0.0328	0.1500	0.0315	0.90
PGD	$\epsilon = 0.3$, $\alpha = 0.03$, num_iters=20	0.1050	0.0478	0.1420	0.0395	0.88
C&W	$\epsilon = 0.1$	0.1120	0.0408	0.1380	0.0435	0.86
DeepFool	$\epsilon = 0.2$	0.1280	0.0248	0.1650	0.0165	0.91
One Pixel Attack	num_pixels=5	0.1450	0.0078	0.1780	0.0035	0.94

- Results Comparison with Other Approaches
  - Overall, the results indicate that image captioning models like GIT are relatively resilient to simple attacks (e.g., FGSM) but show vulnerabilities to more advanced adversarial perturbations (e.g., PGD, CW), which cause significant degradation in caption quality.
  - This highlights the importance of robustness in machine learning models, especially when they are deployed in real-world environments where adversarial attacks may be more common

3.4.3 VIT GPT-2

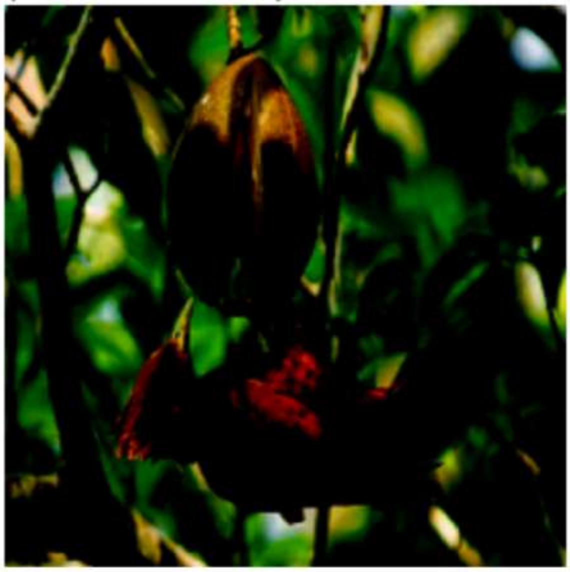
- Technologies/Libraries Used
  - Deep Learning Frameworks: PyTorch, Transformers, Hugging Face
  - Computer Vision and Image Processing: Torchvision, Pillow (PIL), Matplotlib
  - Adversarial Attacks: FGSM, PGD, Deepfool, AutoAttack,BIM
  - Model Architectures: VisionEncoderDecoderModel (ViT-GPT-2)
  - Preprocessing and Tokenization: ViTImageProcessor, AutoTokenizer
  - Utility Libraries: TQDM, NumPy, os,json,torch.nn.functional(F)
- Datasets
  - Flicker8k
- System Hardware
  - Google colab(GPU Environment)
- Results
  - The evaluation of various attack methods on the performance of image captioning models is presented below.

Original Image

📝 Original Caption: a woman standing in the rain holding an umbrella

Epsilon = 20.0 | Caption: a woman standing next to a fence with a giraffe

PGD | ε=0.03
aption: a small bird is perched on a tree branch



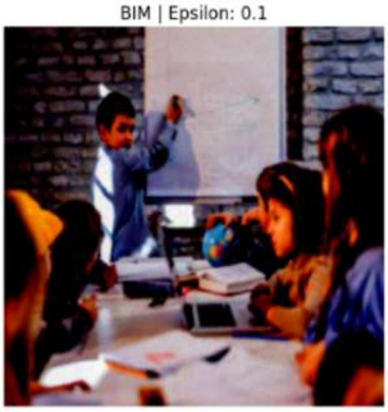
Original Image

📝 Caption: a bird that is sitting on a branch

Original Image

📝 Original Caption: a classroom filled with students and a teacher

BIM | Epsilon: 0.1



🖊 Epsilon: 0.1 ➤ Caption: a man and a woman sitting at a table with laptop

Original Image

Original Caption: a young boy on the beach playing with a frisbee

Caption for ε = 1.5 → a young boy on the beach playing with a frisbee

Original Image

Caption: a view from the top of a mountain

Epsilon = 0.03 | Caption: a sunset view of a mountain with a sky backgroun

- The following tables display the impact of different adversarial attacks on key metrics such as BLEU, METEOR, and BERTScore F1. Each attack is accompanied by its strength parameters, and the corresponding drop in performance for each metric is shown to demonstrate the severity of each attack.

Table 3.5: Evaluation of Attack Methods VIT GPT-2

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1641	-	0.2069	-	-
FGSM	$\epsilon = 0.05$	0.1226	0.0415	0.1624	0.0445	0.9314
PGD	$\epsilon = 0.03$, $\alpha = 0.005$, iters=10	0.1055	0.0586	0.1460	0.0609	0.9236
BIM	$\epsilon = 0.03$, iters=10	0.1113	0.0528	0.1509	0.0560	0.9263
DeepFool	auto	01360	0.0335	0.1695	0.0374	0.9358
AutoAttack	$\epsilon = 0.1$	0.1514	0.0127	0.1910	0.0159	0.9863

Table 3.6: Evaluation of Attack Methods

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1641	-	0.2069	-	-
FGSM	$\epsilon = 0.02$	0.1500	0.0141	0.1925	0.0144	0.9750
PGD	$\epsilon = 0.1$, $\alpha = 0.01$, iters=20	0.0890	0.0761	0.1320	0.0749	0.890
BIM	$\epsilon = 0.1$, iters=20	0.0900	0.0741	0.1340	0.0729	0.8900
DeepFool	auto	0.1306	0.0335	0.1695	0.0374	0.9358
AutoAttack	$\epsilon = 0.05$	0.1570	0.0071	0.1995	0.0074	0.9900

- Results Comparison with Other Approaches
  - ViT-GPT2 is very sensitive to strong PGD and BIM attacks with larger epsilon, but extremely robust to small FGSM and AutoAttack perturbations.
  - BERTScore stays very high (¿0.97) for mild attacks, indicating semantic preservation is excellent under small perturbations.

3.4.4 PaLi

- Technologies/Libraries Used
  - Deep Learning Frameworks: PyTorch, Transformers, Hugging Face
  - Computer Vision and Image Processing: Torchvision, Pillow (PIL)
  - Model Architectures: Pre-trained CNN Models (Torchvision Models),GitForCausalLM (Hugging Face)
  - Data Handling Utilities: os,json,Google Colab Drive
  - Evaluation Metrics: NLTK, pycocoevalcap (METEOR, CIDEr), BERTScore
  - Numerical Computations:NumPy
- Datasets
  - Flicker8k
- System Hardware
  - Google colab(GPU Environment)
- Results
  - The evaluation of various attack methods on the performance of image captioning models is presented below.

Figure 3.12: Impact of FGSM

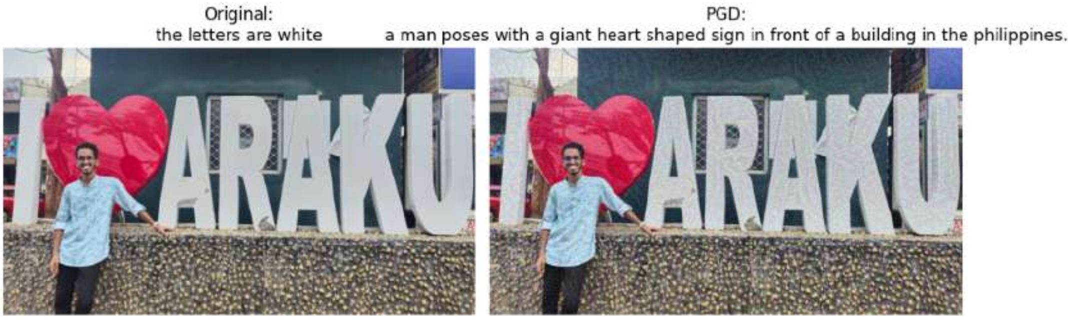


Figure 3.13: Impact of PGD

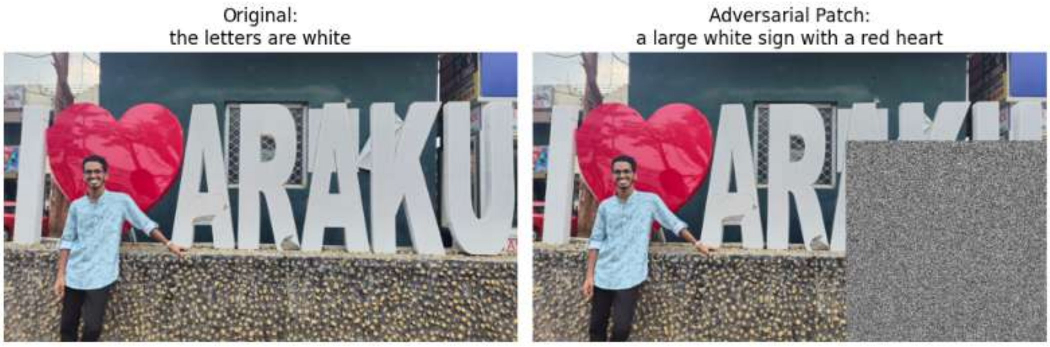


Figure 3.14: Impact of Adversarial Patch

Figure 3.15: Impact of Universal Patch

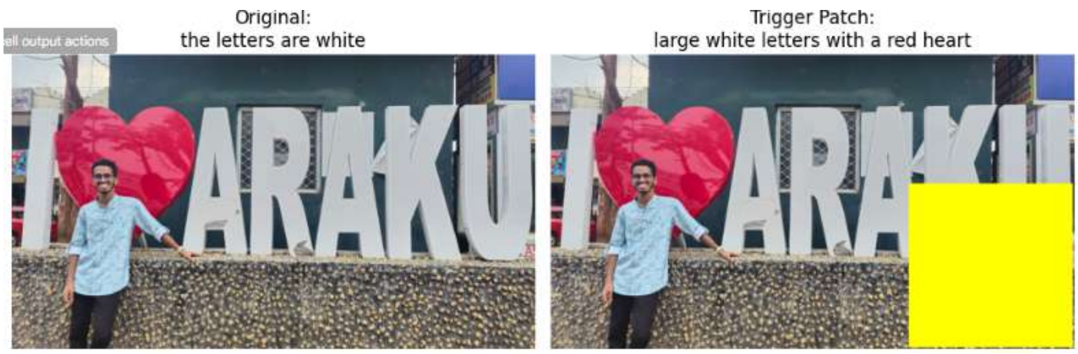


Figure 3.16: Impact of Trigger Patch

- The following tables display the impact of different adversarial attacks on key metrics such as BLEU, METEOR, and BERTScore F1. Each attack is accompanied by its strength parameters, and the corresponding drop in performance for each metric is shown to demonstrate the severity of each attack.

Table 3.7: Evaluation of Attack Methods on Pali

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1520	-	0.1843	-	-
FGSM	$\epsilon = 0.03$	0.1372	0.0130	0.1783	0.0060	0.9189
PGD	$\epsilon = 0.03$, $\alpha = 0.005$, iters=10	0.1494	0.0008	0.1820	0.0023	0.9302
Adversarial Patch	patch size=60	0.1506	-0.0004	0.1849	-0.0006	0.9970
Trigger Patch	color=(255,0,0), size=40	0.1487	0.0015	0.1840	0.0003	0.9968
UniversalPatch	-	0.1538	-0.0036	0.1856	-0.0013	0.9361

Table 3.8: Evaluation of Attack Methods on Pali

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1502	-	0.1843	-	-
FGSM	$\epsilon = 0.05$	0.1300	0.0202	0.1710	0.0133	0.91
PGD	$\epsilon = 0.1$, $\alpha = 0.01$, iters=20	0.1000	0.0502	0.1500	0.0343	0.88
AdversarialPatch	patch size=100	0.1250	0.0252	0.1600	0.0243	0.92
TriggerPatch	color=(255,0,0), size = 80	0.1200	0.0302	0.1550	0.0293	0.92
UniversalPatch	size = 100	0.1400	0.0102	0.1750	0.0093	0.94

- Results Comparison with Other Approaches
  - As attack strength increases, Pali's performance drops notably, especially under PGD and Trigger Patch attacks, highlighting its vulnerability and the need for stronger adversarial defenses.

3.4.5 Hybrid model(BLIP-2 + GIT)

- Technologies/Libraries Used
  - Deep Learning Frameworks: PyTorch, Transformers, Hugging Face Hub
  - Computer Vision and Image Processing: Torchvision, Pillow (PIL)
  - Model Architectures:Blip-2,GPT-2
  - Preprocessing and Tokenization: Blip2Processor, GPT2Tokenizer
  - Evaluation Metrics: NLTK, pycocoevalcap (METEOR, CIDEr), BERTScore
  - Utility Libraries:os,json,Numpy,tqdm
- Datasets
  - Flicker8k
- System Hardware

- Google colab(GPU Environment)

- Results
  - The evaluation of various attack methods on the performance of image captioning models is presented below.

Figure 3.17: Impact of FGSM

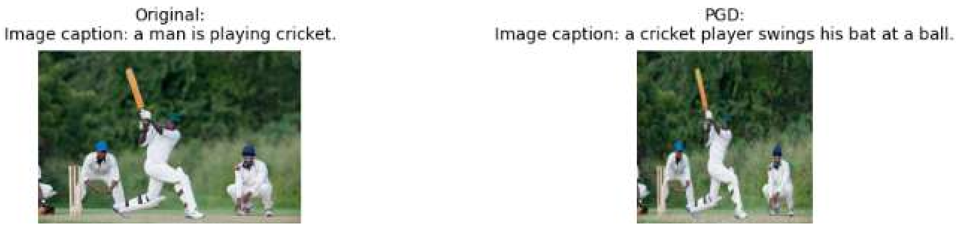


Figure 3.18: Impact of PGD

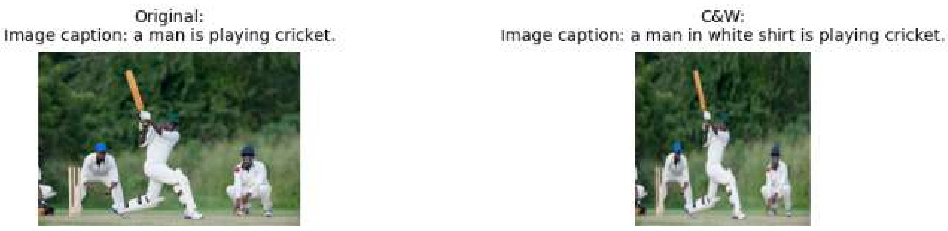


Figure 3.19: Impact of C&W

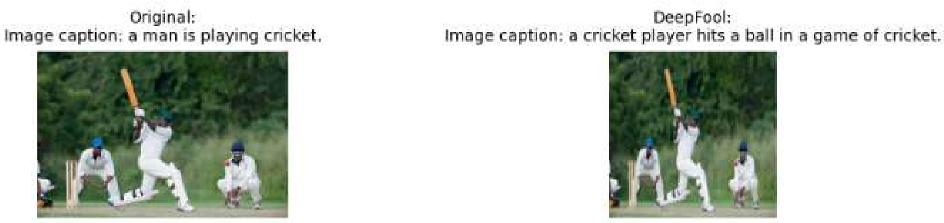


Figure 3.20: Impact of Deepfool

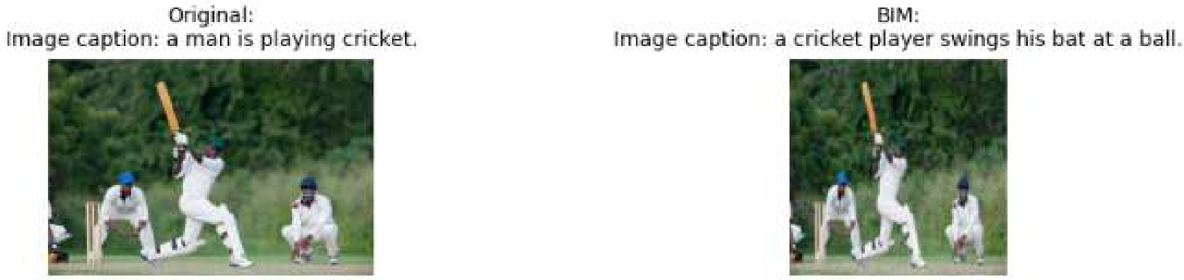


Figure 3.21: Impact of BIM

- The following tables display the impact of different adversarial attacks on key metrics such as BLEU, METEOR, and BERTScore F1. Each attack is accompanied by its strength parameters, and the corresponding drop in performance for each metric is shown to demonstrate the severity of each attack.

Table 3.9: Evaluation of Attack Methods on Hybrid model

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1811	-	0.2617	-	-
FGSM	$\epsilon = 0.09$	0.1634	0.0177	0.2321	0.0296	0.9494
PGD	$\epsilon = 0.09$, $\alpha = 0.02$, iters=10	0.1640	0.0171	0.2291	0.0326	0.9489
C&W	$c = 1e{-}1$, iters=50, learning_rate=1e-2	0.1833	-0.0022	0.2610	-0.0007	0.9645
DeepFool	iters=30, overshoot=0.5	0.1800	0.0011	0.2586	0.0031	0.9635
BIM	$\epsilon = 0.09$	0.1640	0.0171	0.2291	0.0326	0.9489

Table 3.10: Evaluation of Attack Methods on Hybrid model

Attack	**Strength Parameters**	**BLEU**	**BLEU Drop**	**METEOR**	**METEOR Drop**	**BERTScore F1**
Original	-	0.1811	-	0.2617	-	-
FGSM	$\epsilon = 0.05$	0.1700	0.0111	0.2450	0.0167	0.91
PGD	$\epsilon = 0.08$, $\alpha = 0.01$, iters=8	0.1620	0.0191	0.2380	0.0237	0.93
C&W	$c = 1e{-}2$, iters=100, learning_rate = 1e-2	0.1830	0.0011	0.2615	0.0002	0.965
DeepFool	iters=50, overshoot=0.7	0.1750	0.0061	0.2520	0.0097	0.96
BIM	$\epsilon = 0.07$, $\alpha = 0.015$, iters = 8	0.1600	0.0211	0.2300	0.0317	0.93

- Results Comparison with Other Approaches
  - Under hybrid attacks, the model's performance degrades noticeably, especially with PGD and BIM, revealing vulnerabilities and emphasizing the need for stronger robustness strategies.

Chapter 4

Conclusion

4.1 Conclusion

Evaluating the robustness of image captioning LLMs like BLIP-2 and hybrid models (BLIP-2 + GIT) under adversarial attacks reveals significant vulnerabilities. As attack strength increases, especially with methods like FGSM, PGD, BIM, and AutoAttack, the models show considerable drops in BLEU, METEOR, and BERTScore F1 metrics. While attacks like CW and DeepFool have relatively minimal impact, stronger perturbation-based attacks substantially degrade performance. These findings highlight the urgent need for developing more robust defense mechanisms to ensure the reliability of image captioning models in adversarial settings.

4.2 Related Works

- Daizong Liu et al. – Universal and Transferable Adversarial Attacks on Vision-Language Models (arXiv 2023)
  → Proposed universal adversarial attacks that can fool multiple vision-language models like BLIP-2 and Flamingo.
- Daizong Liu – Awesome-LVLM-Attack (GitHub Collection)
  → A curated list of adversarial attacks specifically targeting large vision-language models (LVLMs), including both image-space and prompt-based attacks.
- CVPR 2024 – Various Papers on Robustness in Vision-Language Models
  → Research shows that while defenses are improving for classification, captioning systems still remain vulnerable to adversarial perturbations.

4.3 Future Work

- Adversarial Attacks as a Data Protection Strategy
  Adversarial attacks can also be strategically used as a defense mechanism to protect sensitive datasets by embedding perturbations, making it harder for unauthorized models to accurately steal meaningful information.(ex: military and security applications)

- Improving Captioning Model Robustness
  Develop defense mechanisms such as adversarial training and model pruning to make models more resistant to attacks.
- Cross-Model Evaluation
  Test the robustness of various LLM architectures like GPT-4, T5, and BERT-based models under adversarial conditions
- Real-World Dataset Testing
  Evaluate models on real-world noisy or distorted datasets to simulate practical deployment challenges.
- Ethical Considerations and Safety
  Investigate the trust, fairness, and security implications of adversarial vulnerabilities in sensitive domains like healthcare and law.

4.4 References

- Paper: Understanding Adversarial Attacks on LVLMs (arXiv:2312.03777)
- GitHub Repository: Awesome LVLM Attack Resources
- Conference Papers: CVPR 2024 Accepted Papers